



School of Engineering

Mental Illness Prediction Project

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Summary

The growing prevalence of mental health issues highlights the urgent need for predictive tools that support early intervention. This Machine Learning project focuses on forecasting the presence of mental illness in patients using a rich and diverse dataset that encompasses behavioral, social, medical, and demographic variables.

The data originates from the 2019 Patient Characteristics Survey (PCS), a large-scale dataset provided by the New York State Office of Mental Health (OMH). The PCS collects standardized information from public mental health programs licensed or funded by OMH across New York State. The dataset includes almost 200,000 observations and more than 70 features detailing patient characteristics such as age, gender, diagnosis, service access, substance use, chronic conditions, employment, education, and more.

The core objective of this project is to develop a complete Machine Learning pipeline to predict whether a patient is likely to suffer from a mental illness (binary classification task: Mental Illness = YES or NO). The pipeline encompasses all critical stages:

- Data preprocessing and cleaning
- Exploratory Data Analysis (EDA)
- Feature engineering and selection
- Model training, comparison, and evaluation
- Deployment of a web-based prediction interface

Through rigorous data preparation and modeling, the project aims not only to deliver an effective predictive model but also to derive meaningful insights into patient profiles, comorbidities, and behavioral patterns associated with mental illness.

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I- Introduction

This project is based on the 2019 Patient Characteristics Survey (PCS), a comprehensive dataset made publicly available by the New York State Office of Mental Health (OMH). The PCS is conducted every 2–3 years and aims to capture standardized, statewide data about individuals receiving care from public mental health services across New York. It offers a detailed, high-dimensional view of mental health patient profiles in real-world clinical and social contexts.

The dataset includes almost 200,000 rows and 70+ variables describing patient demographics, diagnoses, insurance coverage, employment status, housing conditions, chronic medical problems, behavioral patterns (e.g., alcohol, drug use), and mental health indicators. Most of the variables are categorical, with a few continuous fields.

The central task of the project is to predict the presence or absence of mental illness for each patient, formulated as a binary classification problem, with the target variable Mental Illness encoded as:

- (YES): if the patient shows sufficient evidence of a mental disorder
- (NO): if not

However, the label Mental Illness is not always reliable. It is often derived indirectly or inconsistently based on other diagnosis-related fields, such as Serious Mental Illness, Principal Diagnosis Class, and Additional Diagnosis Class. This label uncertainty introduces a major challenge and requires careful inspection, harmonization, and correction before any model training.

Additionally, the dataset poses several data quality issues: Missing values in key feature, Logical contradictions across diagnosis columns, Strong class imbalance in the target variable : most patients are labeled as YES, leaving few NO cases to learn from

To address these issues and ensure trustworthy learning conditions, the project follows a structured pipeline consisting of:

Data cleaning and formatting, exploratory Data Analysis (EDA), feature engineering and selection, model selection and evaluation, deployment of a web application for interactive predictions.

By combining clinical logic with machine learning, the project not only builds a robust prediction model, but also brings clarity to diagnostic inconsistencies and improves our understanding of the factors associated with mental illness in real-world setting

II-Exploratory Data Analysis (EDA)

1. Patient Type and Geographic Coverage : The vast majority of patients were served in outpatient programs (67.9%), followed by residential (15.4%) and support programs (10.4%). Most services are concentrated in the New York City region (49.3%), far ahead of other regions like Western NY (17.6%) and Hudson River (14.0%). This confirms a strong urban concentration in the data.

2. Age and Gender : The sample is predominantly composed of adults (77.6%), with children accounting for 22.4%. Gender is nearly balanced, with 50.6% female and 49.2% male patients. A small minority (1.8%) identified as transgender.

3. Sexual Orientation and Ethnicity : Most patients identified as straight or heterosexual (74.6%), but the dataset includes a significant portion of LGBTQ+ individuals (7.9%). Regarding ethnicity, 23.2% were Hispanic/Latino, while 74% were not.

4. Racial Distribution: The population is racially diverse : 49.0% white, 26.0% Black, 18.4% other, and 2.8% multiracial. This diversity must be considered carefully during modeling to avoid biases.

5. Living Situation and Household Composition : 78.5% of patients live in private residences, while 16.3% are in other living arrangements. Most patients cohabit with others (57.1%), but a notable 20% live alone.

6. Religion and Veteran Status : Religious identity is mixed : 35.5% belong to a formal religious group, while 35.7% either consider themselves spiritual or have no spiritual affiliation. Only 2.4% of patients were veterans.

7. Employment and Education : The employment data highlights vulnerability: 64.1% are not in the labor force, while only 17.7% are employed. Regarding education, most patients have at least some high school education, with 51.1% reaching middle or high school, and 27% completing some or full college.

8. Disabilities and Developmental Disorders : Intellectual disabilities are present in 7.7% of patients, Autism spectrum disorders affect 2.7%, Other developmental disabilities (e.g., hearing, visual, speech impairments) are less common, each under 6% prevalence.

This confirms a relatively low incidence of neurodevelopmental conditions in the general dataset.

9. Substance Use and Addiction : Drug-related disorders are present in 14.3% of patients, Alcohol-related disorders in 9.9%, Tobacco use is reported by 25.6%.

10. Chronic Medical Conditions : Conditions such as obesity (15.2%), diabetes (11.0%), and high blood pressure (18.2%) are common. Patients without any chronic condition represent 45.2% of the sample.

11. Mental Health Diagnosis : Serious Mental Illness was reported for 91.7% of patients, suggesting a highly imbalanced target population, Principal Diagnosis Class confirms that 90.8% of patients were diagnosed with "Mental Illness", reinforcing the imbalance, Additional Diagnosis Class shows more variation, with only 34.7% labeled as "Mental Illness", and 30.6% having no additional diagnosis.

III- Methodology

1. Data Cleaning

Before modeling, we performed extensive cleaning of the dataset to address inconsistencies and remove irrelevant features.

- Columns dropped:
 - Survey Year: all values were equal to 2019 and did not bring any predictive value.
 - Three Digit Residence Zip Code: too granular, non-standardized, and possibly privacy-sensitive. It also introduced high cardinality with no added modeling benefit.
 - Preferred Language: removed later due to high dominance of one value (89% English), low variability, and strong correlation with region.
- Standardization of missing values: We replaced inconsistent or incorrectly encoded null values with "np.nan"
- Handling Boolean Columns: All boolean-type columns with missing values were filled with False to indicate absence of the condition or event.

2. Correction of the Target Variable ('Mental Illness')

One of the most critical steps in the pipeline was correcting the Mental Illness label. In the raw data, this column contained multiple inconsistencies:

- Some rows were marked as Mental Illness = YES without any supporting information in the three diagnosis columns.
- Others were marked as NO despite having a valid diagnosis.

To ensure internal consistency and improve label quality, we used the three diagnosis-related columns as cross-check references:

- Serious Mental Illness
- Principal Diagnosis Class
- Additional Diagnosis Class

We applied the following logic:

3-Label Correction Logic:

- If none of the three diagnosis-related columns supported the label "YES", we reclassified it as Mental Illness = 0.
- Additionally, a final harmonization block was applied at the end of the preprocessing pipeline to ensure consistency across the three columns, correcting all remaining NaN and misaligned values.

4- Handling Missing Values (Feature Engineering Strategy)

Our dataset presented a significant number of missing values across multiple columns. To impute them logically and consistently, we followed category-specific strategies.

Children-specific strategy:

Because children were not expected to have jobs or substance disorders, we filled relevant columns as follows:

Financial and Social Help Columns:

Columns such as SSI Cash Assistance, Veterans Disability Benefits, etc., were cleaned of malformed values (like [NO, YES]) and imputed using the mode of each column:

Medical Logic:

- The column No Chronic Med Condition was logically imputed based on the presence of any chronic condition. If at least one condition among 18 was marked "YES", then No Chronic Med Condition = NO.
- Similarly, behavioral fields such as Smokes or Drug Substance Disorder were updated if other treatment indicators were present (e.g., received smoking counseling).

5. Encoding & Harmonization of Categorical Variables

No numerical encoding was applied at this stage, as modeling and vectorization were handled separately. However, we harmonized textual entries to uppercase and trimmed spaces to standardize categories:

a. Final Integrity Checks

We verified consistency between the 3 diagnostic columns and the target variable using harmonization rules such as:

- If Principal or Additional Diagnosis is "MENTAL ILLNESS" : then Mental Illness = 1
- If both are "NOT MI ..." then Mental Illness = 0
- If Mental Illness = 1 but others are missing : we filled the diagnosis fields accordingly

These corrections were essential to ensure the model learns from accurate and consistent data.

IV-Feature Engineering & Selection

1-Extract Missing Values Imputation Strategy

One of the most critical parts of the data preparation process involved managing missing values, which were widespread and heterogeneous across the dataset. Given the complexity and the categorical

nature of most variables in the PCS dataset, we did not opt for a simplistic approach like global mode or median imputation. Instead, we implemented a multi-step, logic-based strategy combining:

- Targeted group-based imputations using conditional modes
- Cascading imputation fallbacks
- Domain-specific assumptions (especially for children)
- Manual logical corrections

a. Grouped Mode Imputation Based on Predictors

We first used a grouped imputation strategy where missing values were filled by the most frequent value (mode) within meaningful subgroups of the population. This allowed us to preserve the distributional patterns specific to each subgroup and avoid introducing global bias. To impute missing values in the variable Sexual Orientation, we grouped by Age Group, Sex, and Region Served, then filled missing entries with the dominant response within that group

Each variable was iterated through this logic using a custom function that:

- Identified valid (non-missing) values within each group
- Retrieved the most frequent value per group
- Assigned it to missing values within that same group

This filled tens of thousands of missing values across critical fields like:

- Transgender (15,426 values filled)
- Religious Preference (56,347 values filled)
- Living Situation, Household Composition, etc.

b. Cascading (Backoff) Imputation for Residual NaNs

For variables where the first-level grouped imputation left residual missing values (due to rare or undefined group combinations), we applied a cascading strategy, also known as backoff imputation.

This means we attempted several grouping combinations in decreasing order of granularity, stopping as soon as the missing values were resolved.

Example : Cascading Imputation for Obesity:

- First, we tried imputing based on the mode in the triple group (Age Group, Sex, Diabetes)
- If not enough data existed in that group, we backed off to (Age Group, Sex)
- Finally, we tried simply Age Group if no other context was usable

This strategy was extremely effective. For example:

- Obesity: 14,376 missing values fully resolved

- Race: 2,685 missing values imputed
- Cannabis Medicinal Use: 7,758 values imputed after fallback

This method allowed us to fill residual NaNs without arbitrarily defaulting to global values.

b. Logical Imputation for Children :

Domain knowledge informed us that children (Age Group = 'CHILD') were unlikely or ineligible to report certain adult-specific attributes. We therefore made several assumptions to directly fill missing values for these subpopulations.

Addiction-related variables such as Alcohol Disorder, Drug Substance Disorder, Smokes, Cannabis Use were set to 'NO' for children with missing values, under the assumption that such conditions were improbable or unrecorded in that age group.

Similarly:

- Employment Status and Number Of Hours Worked were set to 'NOT APPLICABLE'
- Insurance fields were filled accordingly: Child Health Plus Insurance = 'YES', others = 'NO'
- Identity-related fields (Transgender, Sexual Orientation, Religious Preference) were given default values

These rules ensured data completeness while respecting medical and social logic.

2- Encoding :

a. Ordinal Encoding:

In our dataset, some categories follow a hierarchical order. For these variables, we used ordinal encoding, assigning each category a number that reflects its rank and relative importance within the group.

- Living Situation: The categories are ranked by the patient's level of autonomy (from less autonomous to independent) which can give an indication of their mental and socio-economic condition.

- Education Status: The reasoning here is that education levels progress logically from no schooling to higher education, it influences a person's ability to process information, access care, and secure better job opportunities.

- Employment Status: Ranges from inactivity to unpaid work to paid employment, reflecting the engagement and integration into the employment sector.

- Age Group: Even with only two categories (child, adult), the order matters as age influences autonomy, decision-making, and social responsibilities.

- Number of Hours Worked Each Week: A categorical representation of work hours ranges in ascending order, where more hours generally mean higher income and an impact on lifestyle.

b. One-Hot Encoding:

In our dataset, many variables are nominal, meaning their categories represent distinct labels without any order or ranking. Encoding them ordinally would create a false hierarchy and introduce bias into the models.

To preserve the neutrality of these categories and ensure that each label is treated independently, we applied one-hot encoding on the following columns:

- *“Region Served”, “Program Category”, “Race”, “Sex”, “Transgender”, “Household Composition”, “Sexual Orientation”, “Religious Preference”, “Additional Diagnosis Class”, “Principal Diagnosis Class”*

c. Binary Variable Transformation :

Many variables in the dataset were initially expressed as Boolean (True/False) or binary values (Yes/No). While Logistic Regression requires numerical inputs, the scikit-learn implementation of Decision Tree and Random Forest also processes only numeric data. Therefore, we encoded 'Yes' and 'True' as **1**, and 'No' and 'False' as **0**.

This transformation ensured full compatibility with our models, established a uniform representation across binary values, and facilitated a clear interpretation of the results.

d. Special Case: the “Medicaid Managed Insurance” column:

For this column, “Not applicable” was treated as equivalent to “No” since both indicate the absence of active insurance. Grouping them before binary encoding removed ambiguity and ensured a coherent representation of binary variables across the dataset.

3- Correlation matrix analysis:

We analyzed the “Pearson correlation matrix” to identify variables strongly related to the column *“Mental Illness”* which is our target variable, and to detect potential highly correlated predictors. In this analysis, we considered the correlation coefficient ν as:

$\nu \geq 0.50$: **strong** correlation, $0.30 \leq \nu \leq 0.49$: **moderate** correlation, $\nu < 0.30$: **weak** or **no correlation**, and **high negative correlation** as $\nu \leq -0.60$.

a. Results with target = Mental Illness:

Two variables are strongly tied to the target:

- Most features show **weak** linear correlations with *Mental Illness* (mostly < 0.1), except for *Serious Mental Illness* (**0.73**).
- **Serious Mental Illness ($\nu = 0.73$, strong positive)**: It is closely tied to “Mental Illness” as a **subcategory**, making it redundant and risking label leakage (when the model learns from

information that gives away the target), which leads to high training accuracy but worse results in real use.

- **Principal Diagnosis Class NOT MI ($v=-0.617$, strong negative)** : The strong negative correlation between “Principal Diagnosis Class NOT MI (Other)” and the “Mental Illness” indicates that being in a non–mental illness diagnostic category is strongly associated with not having a mental illness, which matches the meaning of this category.
- Key variables like **Education Status**, **Program Category**, or **Household Composition** have limited direct influence, which explains why we are using models like Random Forest to capture non-linear patterns.

b. Strong Predictor Correlations:

Many predictors are highly correlated, showing redundancy:

- “**Household Composition (NOT APPLICABLE)**” and “**Living Situation**” ($v = -0.95$)
- “**Special Education Services (NOT APPLICABLE)**” and “**Age Group**” ($v = 0.93$)
- “**Employment Status**” and “**Number of Hours Worked Each Week**” ($v = 0.90$)
- “**Medicare Insurance with Medicaid**” and “**Medicare Insurance**” ($v = 0.83$).

These high correlations suggest redundancy among predictors, requiring variable selection, variable grouping (merging), or the application of methods to reduce multicollinearity.

c. Decision to Preserve All Predictors:

All variables were retained to keep the dataset’s full predictive potential. While some predictors are strongly correlated, no features were dropped, as the modeling approach (like Random Forest) is robust to multicollinearity, and potential redundancy can be easily managed during feature importance analysis.

4. Predictive Model Selection

Given our clinical context and the need for both dependable predictions and transparent reasoning, we chose to evaluate Random Forest, Logistic Regression, and Decision Tree Classifier. Each offers a different balance of accuracy and interpretability, making them suitable for healthcare data and for producing results that clinicians can easily understand, trust and use.

a. Logistic Regression :

- Provides clear coefficients for each one-hot encoded feature (like the column “**Education Status**” or “**Program Category**”), making it easy to interpret how each category impacts the predicted outcome.
- Performs well when the input features have a linear relationship with the target, meaning that changes in categories like “**SSI Cash Assistance**” lead to consistent and proportional changes in the probability of predicting the column “**Mental Illness**”.

b. Random Forest:

- Handles complex, non-linear relationships like the columns **“Program Category”**, **“Education Status”**, and **“Obesity”** that interact with the features **“High Blood Pressure”** or **“Housing Status”** to affect mental illness.
- Works well with **weakly correlated** features like **“Obesity”** or **“Smoking Counseling”**.
- Highlights key predictors such as **“Serious Mental Illness”**, **“SSI Cash Assistance”**, and **“Education Status”**.

c. Decision Tree Classifier:

- Splits data with clear rules, (if **“Serious Mental Illness”** is **“Yes”**, then we check **“Program Category”** or **“Housing Status”**).
- Highly interpretable, showing exactly how combinations of features lead to a mental illness prediction.

V- Choosing the Best Models for Prediction

After analyzing the results of each of the three models, we chose to select **Random Forest** and **Logistic Regression** as final models.

The **Decision Tree** showed good results overall but has a lower performance than the two others especially on the **minority class “Class 0”** (precision: 0.79, recall: 0.81), resulting in more false positives (predicting mental illness in individuals without it).

Its tendency to overfit reduced its **reliability** for **real-world** clinical applications.

1- Comparison and Evaluation :

We evaluated **Logistic Regression** and **Random Forest** using key metrics (Accuracy, F1-score, precision, recall) to measure how well each model predicts mental illness.

The comparison was done first **without SMOTE**, then **with SMOTE** applied, to observe the impact of balancing the highly imbalanced dataset. The tables below summarize the performance before and after using SMOTE.

a. Before using SMOTE :

Class 0: Minority Class

Class 1: Majority Class

Metric	Accuracy	F1 – score	F1 – score	Precision	Precision	Recall	Recall
Model		Class 0	Class 1	Class 0	Class 1	Class 0	Class 1
Logistic Regression	0.97	0.73	0.98	0.57	1.00	1.00	0.96
Random Forest	0.98	0.83	0.99	0.89	0.99	0.77	1.00

- Random Forest performed better than Logistic Regression overall, especially in the minority class (Class 0), with higher **recall** and **F1-score**. It offers a better balance, making it more suitable for predicting both classes in an **imbalanced** dataset.

b. What Is SMOTE :

SMOTE (Synthetic Minority Over-sampling Technique) addresses class imbalance by generating synthetic minority-class samples through interpolation, enhancing a model's ability to learn decision boundaries.

In our dataset (97% 'Yes', 3% 'No'), this imbalance limited minority-class performance. We applied SMOTE to improve class balance, and among all classifiers tested, Random Forest and Logistic Regression achieved the highest performance.

Metric	Accuracy	F1 – score Class 0	F1 – score Class 1	Precision Class 0	Precision Class 1	Recall Class 0	Recall Class 1
Model		With SMOTE	With SMOTE	With SMOTE	With SMOTE	With SMOTE	With SMOTE
Logistic Regression	0.97	0.75	0.98	0.60	1.00	0.99	0.97
Random Forest	0.98	0.78	0.99	0.82	0.99	0.74	0.99

With **SMOTE**, Random Forest remains the best model overall, with improved performance in **Class 1**. However, its performance on **Class 0** slightly drops compared to before using **SMOTE**, showing a compromise to achieve better class balance.

2 - Results:

After applying SMOTE, **Class 1** is better detected, while **Class 0** sees a slight drop in performance (F1-score from 0.83 to 0.78, precision from 0.89 to 0.82) due to the model focusing more on balance. Despite this, **Random Forest** remains the best model, with strong performance in both classes, good handling of complex data, and robustness to feature interactions. We therefore selected **Random Forest** as the **final model**.

VI. Deployment of the Predictive Model as a Web Application:

To make the predictive model accessible and user-friendly, we deployed the selected **Random Forest** classifier as an interactive web application using **Streamlit**. This interface allows users to upload a CSV file containing patient data and instantly receive predictions on the risk of mental illness. The application loads the trained Random Forest model and applies the same preprocessing steps used during training. This includes cleaning placeholder values, imputing missing data, encoding

categorical features (ordinal and one-hot), converting binary responses to 1 or 0, and aligning all features to the expected format. Once prepared, the data is passed to the model, and predictions are displayed in the interface, ensuring consistent and reliable results.

VII-Conclusion and Recommendations for Improvement

This project successfully developed a full end-to-end Machine Learning pipeline to predict the presence of mental illness using the 2019 Patient Characteristics Survey (PCS) dataset from the New York State Office of Mental Health. Through rigorous preprocessing, label correction, and logic-driven handling of missing values, we ensured that the data used for modeling was both consistent and clinically meaningful. The combination of grouped-mode imputations and domain-specific rules, particularly for children, helped preserve population-specific patterns while minimizing bias introduced during imputation.

Exploratory Data Analysis (EDA) revealed valuable insights into the demographic, clinical, and social composition of the population served by public mental health programs. These findings highlighted the **strong prevalence** of urban service concentration, the **diversity of racial and ethnic backgrounds**, the **socioeconomic vulnerabilities**, and the high rates of comorbid chronic conditions and addictions in the dataset. Importantly, the target variable *“Mental Illness”* was found to be highly imbalanced and occasionally inconsistent, requiring careful harmonization with related diagnostic columns to prevent misleading model learning.

Our model comparison demonstrated that Random Forest consistently outperformed Logistic Regression and Decision Tree Classifier, especially in handling the severe class imbalance. The use of SMOTE improved minority-class detection without excessively compromising majority-class performance. Ultimately, the Random Forest model was selected for deployment because of its ability to capture complex non-linear relationships, robustness to correlated predictors, and balanced performance across both classes.

The deployment of the final model via a Streamlit web application makes the system accessible and practical for real-world use. By automating data preprocessing and ensuring that all transformations applied during training are replicated for new inputs, the application provides consistent, reproducible, and interpretable predictions.

From a broader perspective, the methodology developed in this study provides a reusable blueprint for predictive analytics in healthcare, particularly when working with heterogeneous, categorical-heavy, and imbalanced datasets. The results show that while certain features (such as *“Serious Mental Illness”*) have strong predictive power, others that are traditionally associated with mental health outcomes, like *“Drug Substance Disorder”*, show surprisingly weak correlations in this dataset, suggesting that feature grouping or interaction terms may be more informative than individual predictors.

Recommendations for future work include:

- **Handle High-Leakage Variables**
Remove or mask columns such as *“Serious Mental Illness”* to better assess generalization.

- **Create Aggregated Indices**
Combine related variables into thematic scores. For example, a **Substance Use Index** could merge features such as *"Alcohol Related Disorder"* and *"Opioid Related Disorder"* to capture the combined effect of multiple substance-related conditions. Similarly, a **Chronic Condition Burden score** (including conditions such as *"Obesity"*, *"High Blood Pressure"*, *"Stroke"*, etc.) could summarize the presence of several chronic illnesses into a single metric.
- **Incorporate Temporal Data**
Integrate date-stamped variables, such as *"Alcohol 12m Service"* or *"Diabetes Diagnosis"*, to enable longitudinal tracking of patient status and forecasting of future health outcomes.
- **Test Gradient Boosting Models**
Evaluate *XGBoost* or *LightGBM* to capture non-linear interactions and detect rare patterns, such as *"Traumatic Brain Injury"* with *"Liver Disease"*, or specific *"Age Group"*, *"Principal Diagnosis Class"* combinations.

Suggestions for Handling missing Data

- Use predictive models (e.g., *Random Forest*) to fill important gaps, such as in *"Education Status"* or *"Region Served"*.
- Add missingness indicators (yes/no flags) for columns like *"Alcohol Related Disorder"* or *"Medicaid Insurance"*, so the model knows when data was missing.
- Test model results with and without imputed variables (e.g., *"Obesity"*, *"Principal Diagnosis Class"*) to check stability.
- Remove columns with more than 40% missing values if they are not very useful (e.g., *"Veterans Disability Benefits"*).

In conclusion, this project not only demonstrates the feasibility of predicting mental illness with a high level of accuracy but also emphasizes the importance of thoughtful data preparation and model validation in healthcare Artificial Intelligence. By combining statistical rigor with clinical logic, the developed system provides a foundation for scalable, interpretable, and ethically responsible predictive tools in mental health care.

Project Source Code and Resources

The complete source code, data processing scripts, and deployed web application are available in the following GitHub repository:

<https://github.com/Nirusa04/Machine-Learning-Group-Project>

This repository, hosted by a team member, contains the collective contributions of all project participants.